Exercise 3.13

1. Solve by cross-multiplication method

   (i) $8x-3y=12\,;\ 5x=2y+7$ (ii) $6x+7y-11=0\,;\ 5x+2y=13$

   (iii) $\frac{2}{x}+\frac{3}{y}=5\,;\frac{3}{x}-\frac{1}{y}+9=0$

2. Akshaya has 2 rupee coins and 5 rupee coins in her purse. If in all she has 80 coins totalling ₹ 220, how many coins of each kind does she have.

3. It takes 24 hours to fill a swimming pool using two pipes. If the pipe of larger diameter is used for 8 hours and the pipe of the smaller diameter is used for 18 hours. Only half of the pool is filled. How long would each pipe take to fill the swimming pool.

3.8.3 Consistency and Inconsistency of Linear Equations in Two Variables

Consider linear equations in two variables say

$a_1x+b_1y+c_1=0$ ...(1)

$a_2x+b_2y+c_2=0$ ...(2) where a_1, a_2, b_1, b_2, c_1 and c_2 are real numbers.

Then the system has :

(i) a unique solution if $\frac{a_1}{a_2}\uparrow\frac{b_1}{b_2}$ (Consistent)

(ii) an Infinite number of solutions if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$ (Consistent)

(iii) no solution if $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$ (Inconsistent)

Example 3.54

Check whether the following system of equation is consistent or inconsistent and say how many solutions we can have if it is consistent.

(i) $2x-4y=7$, $x-3y=-2$ (ii) $4x+y=3$, $8x+2y=6$ (iii) $4x+7=2y$, $2x+9=y$

Solution

Sl. No	Pair of lines	$\frac{a_1}{a_2}$	$\frac{b_1}{b_2}$	$\frac{c_1}{c_2}$	Compare the ratios	Graphical representation	Algebraic interpretation
(i)	$2x-4y=7$ $x-3y=-2$	$\frac{2}{1}=2$	$\frac{-4}{-3}=\frac{4}{3}$	$\frac{7}{-2}=\frac{-7}{2}$	$\frac{a_1}{a_2}\uparrow\frac{b_1}{b_2}$	Intersecting lines	Unique solution
(ii)	$4x+y=3$ $8x+2y=6$	$\frac{4}{8}=\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{6}=\frac{1}{2}$	$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$	Coinciding lines	Infinite many solutions
(iii)	$4x+7=2y$ $2x+9=y$	$\frac{4}{2}=2$	$\frac{2}{1}=2$	$\frac{7}{9}$	$\frac{a_1}{a_2}=\frac{b_1}{b_2}\uparrow\frac{c_1}{c_2}$	Parallel lines	No solution

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